Genuine Multipartite Entanglement Analysis

Hubbard ring - Convex-roof framework

System parameters

Model	**Hubbard 1D-PBC**
Mode	**uniform**
Sites (L)	**6**
Electrons (Ne)	**6**
Hopping (t)	**1.0**
Interaction (U)	**4.0**
n-body order	**4**
Eigenstate idx	**4 (1-indexed: 5)**

Eigenstate properties

Energy E0	**-2.42291126**
Energy E1	**-2.09275383**
Gap (E1-E0)	**0.330157**
<S^2>	**2.000000**
S	**1.0000**
<Sz>	**0.000000**
Multiplicity	**3.0**
Spin label	**Triplet (S=1)**

Double occupancy

<D> total	**0.471901**
<D> / site	**0.078650**

RDM and GME statistics for this run (4-body clusters)

Hilbert space dim : 2^4 = 16
Number of clusters : 6
Pure clusters : 0
Mixed clusters : 6
Clusters with GME (>0) : 6

Purity gamma = Tr(rho^2) : mean = 0.272209 [min 0.261360 , max 0.284715]
Von Neumann entropy : mean = 1.603886 [min 1.578791 , max 1.636351]
Rank of rho : mean = 14.00 [min 14 , max 14]
C_GME (best estimate) : mean = 0.358514 [min 0.282254 , max 0.434884]

NOTE Analysis performed on eigenstate index 4 (state 5 in 1-indexed convention), not the ground state.

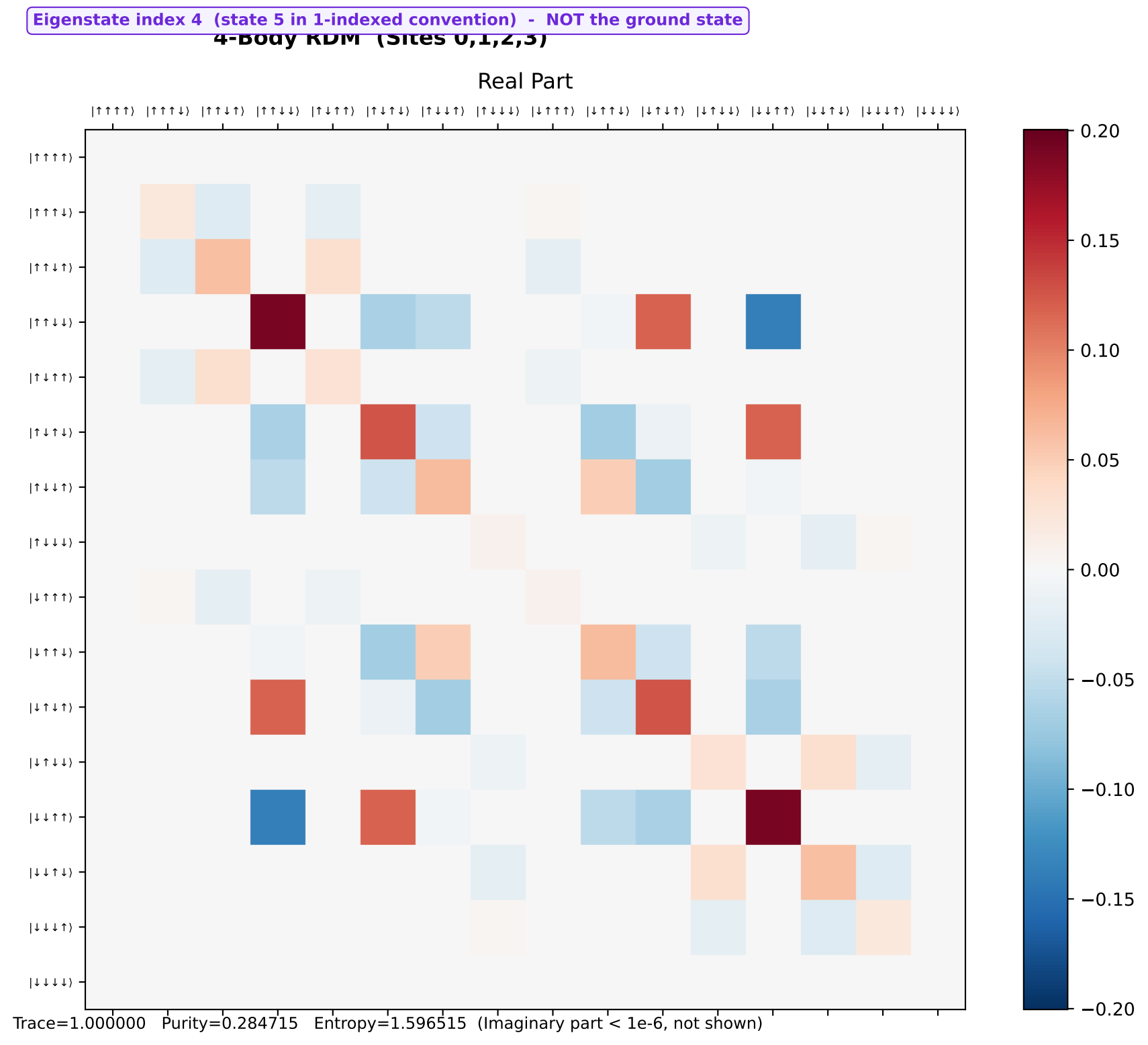
Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state

Summary: All 4-Body RDM Clusters

System: L=6, Ne=6, U=4.0, t=1.0 | E_0=-2.422911

Sites	Purity	C_GME	C_LB	C_cayl	C_expo	C_stiefel	C_UB	Status
(0, 1, 2, 3)	0.284715	–	0.000000	0.000000	0.000000	0.282254	0.555871	MIXED
(1, 2, 3, 4)	0.270551	–	0.000000	0.000000	0.000000	0.427001	0.617949	MIXED
(2, 3, 4, 5)	0.261360	–	0.000000	0.000000	0.000000	0.343927	0.537543	MIXED
(3, 4, 5, 0)	0.284715	–	0.000000	0.000000	0.000000	0.303810	0.555871	MIXED
(4, 5, 0, 1)	0.270551	–	0.000000	0.000000	0.000000	0.434884	0.617949	MIXED
(5, 0, 1, 2)	0.261360	–	0.000000	0.000000	0.000000	0.359205	0.537543	MIXED

Legend:
  Purity = Tr(rho²) [1.0=pure, <1.0=mixed] | C_GME = Genuine multipartite entanglement (pure only)
  C_LB = Lower bound (notes eqs.6-9) | C_cayl = unitary optimization - Cayley Transform |C_expo= unitary optimization - Expoenetial Map | BEST
  C_UB = Upper bound (eigendecomp) | Status: GME =has GME | NO GME=separable | MIXED=mixed state


Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state
4-Body RDM (Sites 0,1,2,3)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
Trace=1.000000 Purity=0.284715 Entropy=1.596515 (Imaginary part < 1e-6, not shown)

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state

Analysis Summary - 4-Body RDM (Sites 0,1,2,3)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.2847150458
  Von Neumann entropy  : 1.5965153852
  Eigenvalue min       : 0.000000e+00
  Eigenvalue max       : 4.456593e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.28471505

  A            | B            | Tr(rho_A^2)
  ____________ | ____________ | ___________

STATUS: MIXED STATE
____________________________________

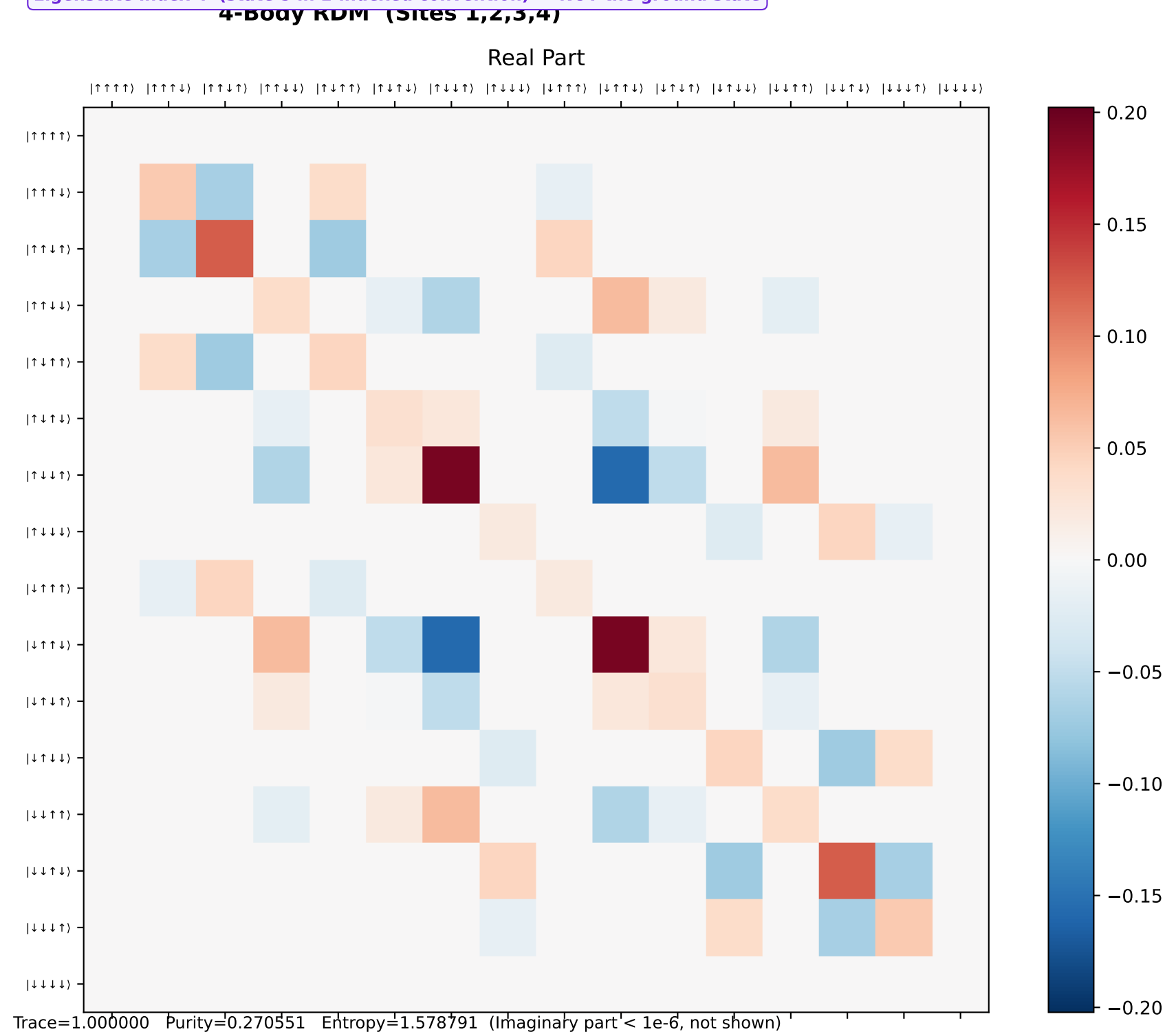
Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.28225388   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.55587119

  Status: nONE
```

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state



Trace=1.000000 Purity=0.270551 Entropy=1.578791 (Imaginary part < 1e-6, not shown)

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state

Analysis Summary - 4-Body RDM (Sites 1,2,3,4)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.2705510517
  Von Neumann entropy  : 1.5787909528
  Eigenvalue min       : 0.000000e+00
  Eigenvalue max       : 4.127248e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.27055105

  A            | B            | Tr(rho_A^2)
  ─────────────┼──────────────┼────────────

STATUS: MIXED STATE
──────────────────────────────────

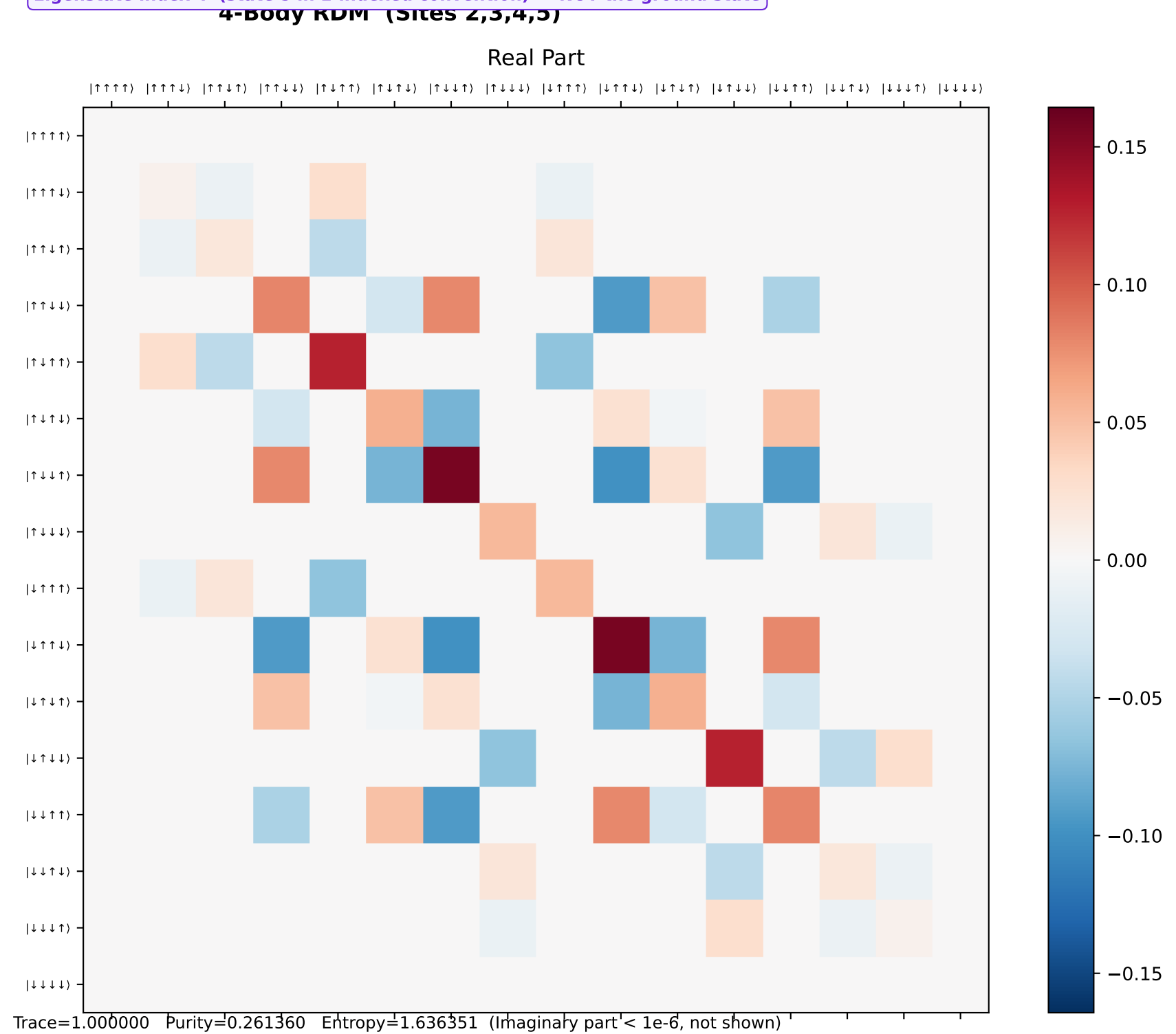
Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.42700140   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.61794910

  Status: nONE
```

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state



Trace=1.000000 Purity=0.261360 Entropy=1.636351 (Imaginary part < 1e-6, not shown)

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state

Analysis Summary - 4-Body RDM (Sites 2,3,4,5)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.2613598576
  Von Neumann entropy  : 1.6363513903
  Eigenvalue min       : 0.000000e+00
  Eigenvalue max       : 4.225758e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.26135986

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

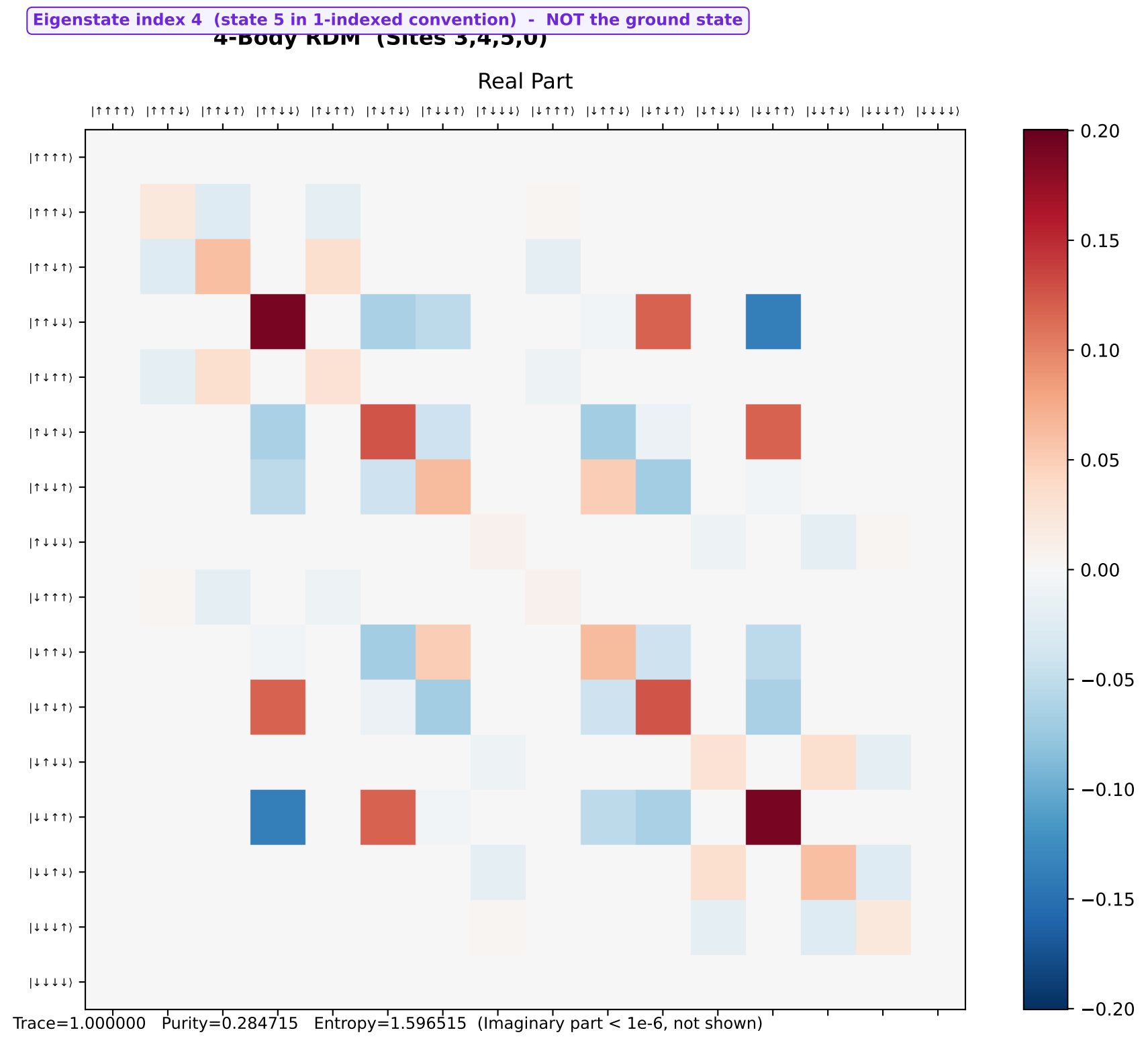
STATUS: MIXED STATE
------------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.34392680   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.53754285

  Status: nONE
```


Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state
4-Body RDM (Sites 3,4,5,0)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
Trace=1.000000 Purity=0.284715 Entropy=1.596515 (Imaginary part < 1e-6, not shown)

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state

Analysis Summary - 4-Body RDM (Sites 3,4,5,0)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.2847150458
  Von Neumann entropy  : 1.5965153852
  Eigenvalue min       : 0.000000e+00
  Eigenvalue max       : 4.456593e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.28471505

  A            | B            | Tr(rho_A^2)
  ─────────────┼──────────────┼────────────

STATUS: MIXED STATE
──────────────────────────────────

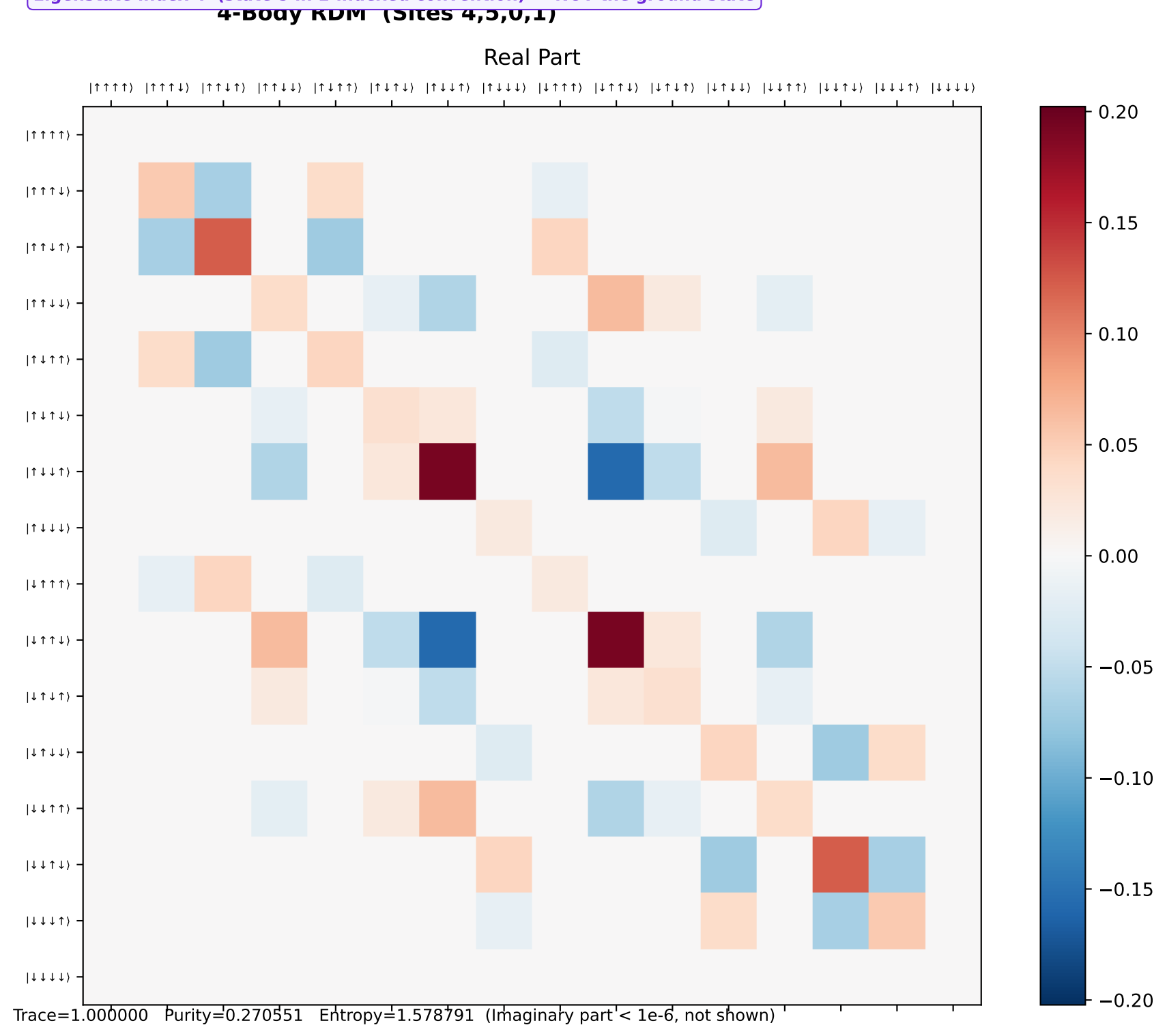
Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.30380970   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.55587119

  Status: nONE
```

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state



Trace=1.000000 Purity=0.270551 Entropy=1.578791 (Imaginary part < 1e-6, not shown)

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state

Analysis Summary - 4-Body RDM (Sites 4,5,0,1)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.2705510517
  Von Neumann entropy  : 1.5787909528
  Eigenvalue min       : 0.000000e+00
  Eigenvalue max       : 4.127248e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.27055105

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
------------------------------------

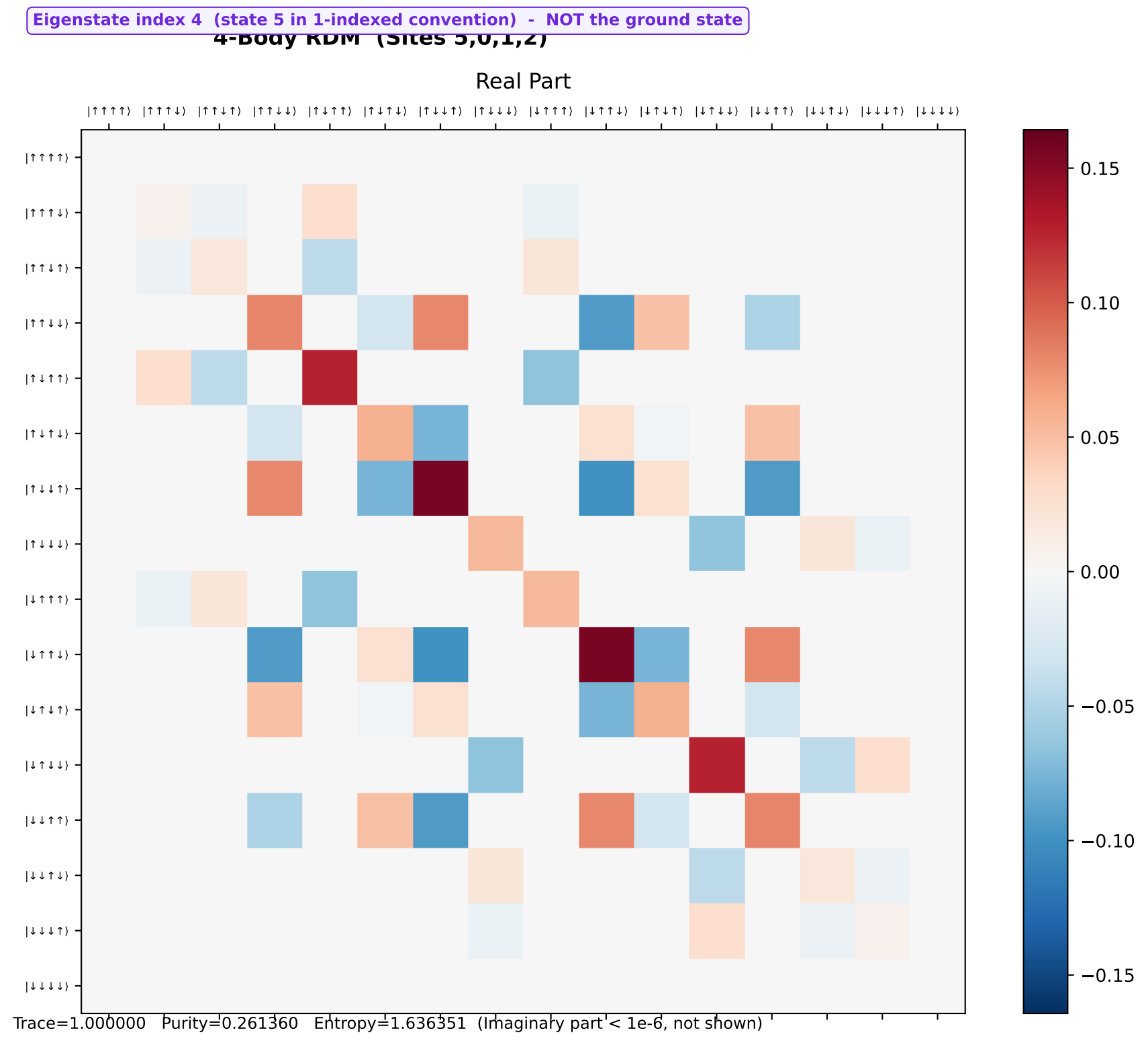
Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.43488413   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.61794910

  Status: nONE
```

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state



Trace=1.000000 Purity=0.261360 Entropy=1.636351 (Imaginary part < 1e-6, not shown)

Eigenstate index 4 (state 5 in 1-indexed convention) - NOT the ground state

Analysis Summary - 4-Body RDM (Sites 5,0,1,2)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.2613598576
  Von Neumann entropy  : 1.6363513903
  Eigenvalue min       : 0.000000e+00
  Eigenvalue max       : 4.225758e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.26135986

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
------------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.35920522   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.53754285

  Status: nONE
```